

Test V2: (1h:30)

Full Name:

Note: Show all the solution steps

Group:

Exercise 1: (1 pts)

Question	Answer
Calculate the maximum value that can be represented by a 10-bit unsigned number?	
What is the smallest number that can be represented using 7 bits in Two's Complement ?	
How many bits are required to represent the decimal number 2047 in binary ?	
How many bits are needed to represent the number -128 in one's Complement ?	

Exercise 2: (1pts)

Determine the base of the numbers in each case for the following operations to be correct:

a. $54 \setminus 4 = 13.$

c. $24 + 17 = 40$

Exercise 3: (6pts)

1. The following conversions contain various types of **mistakes**. Identify the **errors** in each conversion. Provide the correct conversions. show all steps calculation. Take the 1st one as reference and check the others:

- $(25)_{10} = (11000)_2 = (41)_8 = (19)_{16}$
- $(145.625)_{10} = (10011001.100)_2 = (221.5)_8 = (99.4)_{16}$
- $(111001.011)_2 = (57.5)_{10} = (11.3)_8 = (39.6)_{16}$
- $(141.3030)_5 = (101110.100)_2 = (46.625)_{10} = (55.44)_8$
- $(123.3)_4 = (11010.11)_2 = (27.75)_{10} = (55.B)_{16}$
- $(91.5)_{10} = (5B.8)_{16} = (321.3)_6 = (133.4)_8$

Check and correct:

2. Perform the following operations in binary and verify the result using Decimal equivalents.

$(101011.101)_2 + (1110.1011)_2$	$(3E.8)_{16} \times (7.3)_8$
$(1111101.110)_2 - (4F.2)_{16}$	

Exercise 4: (7pts)

1. Complete the table based on the above representations for the given numbers:

Numbers	Decimal	Pure Binary (8-bits)	Sign-magnitude (8-bits)	1's Complement (8-bits)	2's complement (8-bits)
A			1001 1010		
B					0101 1000
C	-10				
D				1001 1011	

2. Using 8-bit representation, Perform the following operations in 2's complement, and put details on each one regarding the carry and the overflow and finally give the decimal equivalents:

<u>A + B</u>	<u>D - B</u>

3. Perform A X D in 8 bits 2's complement using one of the methods presented on the course:

A x C

Good luck